



भारतीय प्रौद्योगिकी संस्थान मुंबई
INDIAN INSTITUTE OF TECHNOLOGY BOMBAY
पवई / Powai, मुंबई / Mumbai 400 076



Roll Number: 200260023
Name of the Student: Kaushik Rajesh Dhandekar
Programme: Bachelor of Technology (B.Tech.)

Academic Unit: Engineering Physics
Joining Month & Year: November 2020

Code	Name	Credits	Tag	Grade/ Marks	Code	Name	Credits	Tag	Grade/ Marks
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Academic Year: 2020 - 2021, Term: Semester Autumn

CH 105 Organic & Inorganic Chemistry	4.0	MA	BC	MA 111 Calculus II	4.0	MA	BC
CH 107 Physical Chemistry	4.0	MA	AA	NOCS01 NCC/NSS/NSO	0.0	MA	PP
CS 101 Computer Programming and Utilization	6.0	MA	BB	PH 107 Quantum Physics and Application	6.0	MA	AB
MA 109 Calculus I	4.0	MA	BB				

SPI=8.21/10

CPI=8.21/10

Academic Year: 2020 - 2021, Term: Semester Spring

BB 101 Biology	6.0	MA	BB	ME 119 Engineering Graphics & Drawing	5.0	MA	AA
EE 112 Introduction to Electronics	6.0	MA	BB	NOCS02 NCC/NSS/NSO	0.0	MA	PP
MA 106 Linear Algebra	4.0	MA	BC	PH 108 Basics of Electricity & Magnetism	6.0	MA	BB
MA 108 Differential Equations	4.0	MA	CC				

SPI=7.94/10

CPI=8.07/10

Academic Year: 2021 - 2022, Term: Semester Autumn

GC 101 Gender in the workplace	0.0	MA	PP	PH 215 Thermal Physics	3.0	MA	BB
GNR401 Remote sensing and Image Processing	6.0	MA	BB	PH 217 Classical Mechanics	6.0	MA	BC
HS 101 Economics	6.0	MA	BB	PH 219 Data Analysis and Interpretation	6.0	MA	CC
MA 205 Complex Analysis	4.0	MA	AB	PH 231 Electronics Lab I (Basic circuits)	3.0	MA	AB
MA 207 Differential Equations II	4.0	MA	BB	PH 233 Electronics Lab II (Op amp circuits)	3.0	MA	BC
PH 207 Introduction to Special Theory of Relativity	3.0	MA	AB				

SPI=7.75/10

CPI=7.93/10

Academic Year: 2021 - 2022, Term: Semester Spring

EE 224 Digital Systems	6.0	MA	BB	PH 204 Quantum Mechanics I	8.0	MA	BC
MA 214 Introduction to Numerical Analysis	8.0	MA	BB	PH 230 Electronics Lab III (Digital Electronics)	5.0	MA	AA
PH 202 Waves & Oscillations & Optics	6.0	MA	BB	PH 232 Physics Laboratory I (General Physics Lab)	3.0	MA	AB

SPI=8.14/10

CPI=7.99/10

Academic Year: 2022 - 2023, Term: Semester Autumn

HS 305 Reading Literature	6.0	MA	BB	PH 423 Quantum Mechanics II	6.0	MA	BC
ME 781 Statistical Machine Learning and Data Mining	6.0	MA	BC	PH 435 Electronics Lab IV (Microprocessors)	5.0	MA	AA
PH 421 Photonics	6.0	MA	BB				

SPI=7.93/10

CPI=7.98/10

Academic Year: 2022 - 2023, Term: Semester Spring

PH 436 Introduction to Condensed Matter Physics	6.0	MA	CC	PH 556 Astrophysics	6.0	MA	BB
PH 438 Statistical Physics	6.0	MA	CC	PH 566 Advanced Simulation Techniques in Physics	8.0	AL	CD
PH 444 Electromagnetic Theory	6.0	MA	BB	PH 578 Nanodevices and Applications	6.0	MA	BB
PH 446 Physics Laboratory (Solid state and Nuclear Physics)	3.0	MA	AA				

SPI=7.45/10

CPI=7.89/10

Academic Year: 2023 - 2024, Term: Semester Autumn

CS 626 Speech and Natural Language Processing and the Web	6.0	AL	BC	PH 505 Introduction to Nuclear & Particle Physics	6.0	MA	BB
HS 636 Development of Mathematics in India	6.0	MA	AB	PH 515 Introduction to Atomic and Molecular Physics	6.0	MA	AB
PH 447 Physics Lab (Optics and Spectroscopy)	3.0	MA	AA	PH 517 Methods in Analytical Techniques	6.0	MA	BC

SPI=8.44/10

CPI=7.96/10

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Name of the Student: Kaushik Rajesh Dhandekar

Roll Number : 200260023

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Academic Year: 2023 - 2024, Term: Semester Spring

ES 200 Environmental Studies: Science and Engineering	3.0	MA	CC	HS 401 Introduction to Paninian Grammar	6.0	MA	AB
GNR640 Geospatial Statistics	6.0	MA	AA	HS 618 Introduction to Indian Astronomy	6.0	MA	AA
HS 200 Environmental Studies	3.0	MA	CC	PH 574 Physics of Semiconductor Devices	6.0	MA	BC

SPI= 8.40/10

CPI= 8.01/10

Mandatory Course Credits (MA)	= 258.0	Overall CPI	= 8.01/10
Overall Credits Completed	= 272.0		
Overall Grade Points	= 2148.0		

Final Result

The student has completed the academic requirements of the programme in the month of May 2024 for the award of Bachelor of Technology in Engineering Physics

Signature & Seal of Transcript Issuing Authority:

[Signature]

Joint/Assistant Registrar (Academic), IIT Bombay

Date: 17-August-2024

Place: ~~पुणे~~ **पुणे** **कुलसचिव (शैक्षिक)**

-----Assistant Registrar (Academic)-----

General ~~पुणे~~ **पुणे** **भारतीय प्रौद्योगिकी संस्थान, मुंबई**

Indian Institute of Technology, Bombay

The medium of instruction at the Institute is English.

Course credits and grade: Each course is associated with credits which are an indicator of its relative weight in calculating the academic performance. A two-letter grade is awarded to students on the basis of their performance in examinations and assignments of a specific course. The letter grades have numerical equivalents on a 0-10 scale as given below.

Letter Grade	AP	AA	AB	BB	BC	CC	CD	DD	FF	FR	W	DX	PP	NP	AU
Numerical Equivalent	10	10	9	8	7	6	5	4	0	0					

FF: Fail, FR: Fail and repeat, W: Withdrawn, DX: Insufficient attendance, AU: Satisfactory performance in an audit course, PP: Pass, NP: Not Pass. The minimum passing grade in a course is DD. The grade AP is awarded to students with exceptional performance in core courses of a programme. Numerical equivalents of letter grades are referred to as grade points.

The numerical grade points are not convertible into marks or percentages.

Performance Indicators: The performance of a student in a semester is given by a number called the Semester Performance Index (SPI), which is the weighted average of the earned grade points in the courses during the semester.

If a student has courses with credits $C_1, C_2, \dots, C_n$, with grade points of $G_1, G_2, \dots, G_n$ respectively, then

Semester Credits = $C_1 + C_2 + \dots + C_n$	Semester Grade Points = $C_1 G_1 + C_2 G_2 + \dots + C_n G_n$	SPI = Semester Grade Points / Semester Credits
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Cumulative Performance Index (CPI) is the weighted average of the grade points in the courses in all semesters. The indices SPI and CPI are calculated upto two decimal places.

Courses are tagged as MA: Mandatory (Core/Elective), MI: Minor, HO: Honours, AL: Additional Learning, AU: Audit

- Each degree programme has mandatory credits consisting of core courses, elective courses, and non credit courses. These courses are tagged as MA.
- For calculation of SPI and CPI, grades obtained only in mandatory courses (MA) are considered.
- Students can supplement the learning experience by crediting additional courses. Credits earned in these courses, when appropriate, can earn additional credentials either in the form of "Honours" (HO) in the chosen discipline or "Minor" (MI) in another discipline or both.
- "Honours" is not indicative of proficiency, and can be earned by completing the additional prescribed set of advanced core and elective courses in the chosen discipline. "Minor" can be earned by completing the prescribed set of courses in a discipline other than the chosen discipline. Additional courses that are not used for earning "Honours" or "Minor" are tagged as "Additional Learning" (AL).
- The AU is awarded based on satisfactory attendance and fulfilling the minimum requirements as set by the course instructor. It carries no grade points and does not figure in SPI or CPI calculations.
- PP or NP is awarded in some credit courses that are not earmarked with a letter grade. Correspondingly, PP/NP does not carry a grade point.
- O-ITB is/are the Course(s) completed by a student outside IIT Bombay (NPTEL/ Swayam/ Semester Exchange). These course(s) contribute towards the completion of credits for a degree requirement. However, grades/marks earned for such course(s) is/are not considered for SPI / CPI calculation.

The Institute does not award any class or division. Notionally, the CPI may be multiplied by a factor of 10 to obtain a numerical percentage.

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END OF TRANSCRIPT

Roll Number: 200260023